

MASTER IN DATA SCIENCE
Complex and Social Networks Laboratory

Network dynamics

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Project Report

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1 Introduction

The spreading of diseases, information, or computer viruses can be studied using epidemic models that describe how individuals change their state over time. One widely used model is the Susceptible-Infected-Susceptible (SIS) model, in which individuals can become infected through contact with infected neighbors and can later recover and become susceptible again. When interactions between individuals are represented as a network, the structure of that network plays an important role in determining how the spreading process evolves.

In this work, we study the SIS model on different network topologies to understand how network structure affects epidemic spreading. In particular, we focus on the role of the leading eigenvalue of the network adjacency matrix, which is known to be related to the epidemic threshold in network-based SIS models. By comparing simulations on networks with different structural properties, we aim to observe how differences in connectivity influence both the persistence and extinction of infections.

The study is divided into two main tasks. In Task 1, we simulate the SIS model using a fixed set of parameters and analyze the time evolution of the proportion of infected nodes for several network types. This allows us to compare how different network structures respond to the same spreading conditions. In Task 2, we compute a network-specific reference value based on the leading eigenvalue and select infection probabilities slightly above and below this value. We then simulate the SIS dynamics to examine how the behavior changes across these two regimes.

Through these experiments, we provide a numerical exploration of SIS spreading on networks and assess how well the simulation results align with the theoretical framework presented in the course material. The results highlight the importance of network topology in epidemic processes and illustrate how spectral properties of networks can be used to characterize their susceptibility to sustained infection.

2 Results

2.1 Task 1

Table 1: SIS simulations with fixed model parameters

Network	λ_1	Inf.% (Mean)	Inf.% (STD)
Erdős–Rényi	11.17	51.81%	1.62%
Barábasi–Albert	20.56	48.52%	1.92%
Watts–Strogatz	6.15	30.72%	2.65%
Complete	999.00	66.69%	1.73%
Star	31.61	17.30%	4.78%

Table 1 reports the leading eigenvalue λ_1 of each network and the mean and standard deviation of the percentage of infected nodes at the final simulation time step. All results correspond to SIS simulations performed with fixed parameters $\beta = 0.15$, $\gamma = 0.5$, and initial infected fraction $p_0 = 0.1$, on networks of size $N = 1000$. The infection percentages are computed from 100 Monte Carlo runs of 50 time steps.

Figure 1 shows the time evolution of the proportion of infected nodes in SIS simulations performed on different network topologies. All simulations use the same model parameters, with infection probability $\beta = 0.15$, recovery probability $\gamma = 0.5$, and initial infected fraction $p_0 = 0.1$, on networks of size $N = 1000$. Each curve represents the mean proportion of infected nodes over 100 Monte Carlo runs, and the shaded regions indicate one standard deviation. The leading eigenvalue λ_1 of each network is reported in the legend.

2.2 Task 2

Table 2 reports the epidemic threshold values computed as the inverse of the leading eigenvalue λ_1 for each network topology. For each network, SIS simulations were performed using parameter values slightly above and slightly below the corresponding threshold. The *Consistent* column indicates whether the final proportion of infected nodes observed above the threshold was greater than that observed below the threshold.

Figure 2 shows the time evolution of the proportion of infected nodes in SIS simulations on different network topologies using parameter values selected relative to the epidemic threshold of each network.

Figure 1: SIS spreading dynamics with fixed model parameters

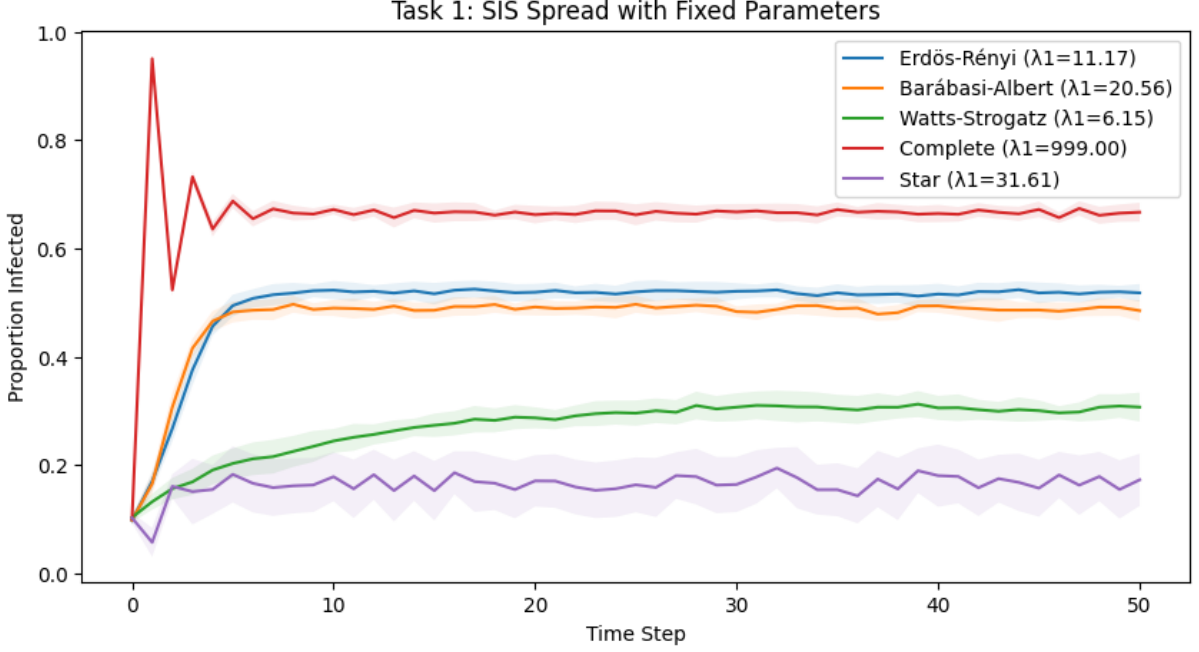


Table 2: estimated epidemic thresholds and simulation consistency

Network	Threshold	Consistent
Erdős-Rényi	0.0896	True
Barabási-Albert	0.0486	True
Watts-Strogatz	0.1625	True
Complete	0.0010	True
Star	0.0316	True

For each network, the threshold is computed as $1/\lambda_1$, where λ_1 is the leading eigenvalue of the adjacency matrix. The left panel corresponds to simulations with infection probability $\beta = 1.3 (\gamma/\lambda_1)$, and the right panel corresponds to simulations with $\beta = 0.7 (\gamma/\lambda_1)$, with recovery probability $\gamma = 0.5$ and initial infected fraction $p_0 = 0.1$. Each curve represents the mean proportion of infected nodes over 100 Monte Carlo runs, and the shaded regions indicate one standard deviation.

3 Discussion

3.1 Task 1

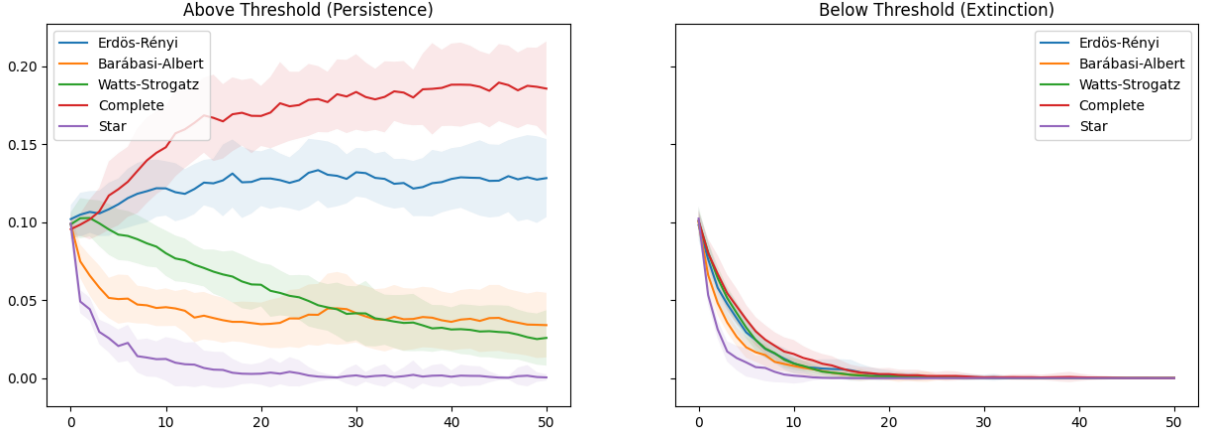
In Task 1, we simulated the SIS model on different network topologies using the same values of the parameters $\beta = 0.15$, $\gamma = 0.5$, and $p_0 = 0.1$. For each network, we measured how the proportion of infected nodes changes over time. The results are shown in Figure 1 and summarized in Table 1 with the final infection levels and the leading eigenvalue λ_1 of each network.

The simulations show that all networks reach a stable level of infection after an initial transient period. However, the final proportion of infected nodes is different for each network. The Complete network shows the highest infection level, followed by the Erdős-Rényi and Barabási-Albert networks. The Watts-Strogatz network reaches a lower infection level, and the Star network shows the lowest average infection prevalence, with larger fluctuations across runs.

From the SIS theory on networks, the epidemic threshold is given by $1/\lambda_1$ is the leading eigenvalue of the network's adjacency matrix. When the parameters β and γ are fixed, networks with a larger λ_1 are expected to be more prone to sustaining the infection, because their epidemic threshold is lower.

This theoretical prediction is reflected in the results. The Complete network has a very large leading eigenvalue ($\lambda_1 = 999$) and also shows the highest steady-state infection level. The Erdős-Rényi and

Figure 2: SIS dynamics above and below the epidemic threshold



Barabási-Albert networks have smaller but still relatively large values of λ_1 , and they show intermediate infection levels. Although the Barabási-Albert network has a larger λ_1 than the Erdős-Rényi network, their infection levels are similar, which may be due to differences in how node degrees are distributed in the two networks.

The Watts-Strogatz network has a smaller leading eigenvalue and correspondingly reaches a lower level of infection. The Star network, even though it has a relatively large λ_1 , shows the lowest average infection prevalence. In this case, most nodes have only one connection, so the spread of the infection depends strongly on the central hub. When peripheral nodes recover, the infection is less likely to persist, which results in lower overall prevalence and higher variability.

In summary, the results show that some networks are more prone to epidemic spreading than others, even when the model parameters are the same. The observed differences in infection prevalence are broadly consistent with the role of the leading eigenvalue λ_1 in SIS epidemic theory

3.2 Task 2

In Task 2, we examined how the SIS dynamics change when the model parameters are selected relative to a characteristic threshold of each network. For each network topology, the leading eigenvalue of the adjacency matrix was first computed, and a corresponding threshold value was obtained as its inverse. This value was used as a reference to guide the choice of simulation parameters.

For each network, two values of the infection probability β were selected: one larger and one smaller than the reference threshold, while keeping the recovery probability fixed at $\gamma = 0.5$ and the initial infected fraction at $p_0 = 0.1$. In the simulations, the infection probabilities were set proportionally to the threshold value, with $\beta_{\text{high}} = 1.3 (\gamma/\lambda_1)$ and $\beta_{\text{low}} = 0.7 (\gamma/\lambda_1)$. This ensured that the two parameter choices were symmetrically placed above and below the reference value for each network.

Using these parameter settings, SIS simulations were run for all network types. The results are summarized in Table 2, which reports the threshold value for each network and whether the simulation outcomes above and below the reference value differ as expected. The corresponding Figure 2 shows the time evolution of the proportion of infected nodes for both parameter choices. Together, the table and the figure allow a direct comparison of the spreading behavior when the infection probability is chosen slightly above or slightly below the network-specific reference value.

The threshold values computed for the networks show clear differences across topologies. The Complete graph has the smallest threshold (about 0.0010), while Watts-Strogatz has the largest (about 0.1625). Erdős-Rényi and Barabási-Albert fall in between, and the Star network has a relatively small threshold as well. These values follow directly from differences in λ_1 , since a larger λ_1 leads to a smaller $1/\lambda_1$.

Figure 2 supports the expected threshold behavior. In the above-threshold panel, each network's infection curve stabilizes at a positive (non-zero) proportion of infected nodes over time. In the below-threshold panel, the infection curves decrease toward values very close to zero as time increases. This pattern appears for all network types tested, meaning the qualitative outcome matches the theoretical prediction from the eigenvalue threshold condition.

This outcome is also reflected in Table 2: the “Consistent” field is True for every network. In the

implementation, this consistency check is based on whether the final average infection proportion for the above-threshold run is larger than the final average infection proportion for the below-threshold run. Since this condition holds for all networks, the simulations are consistent with the threshold theory for SIS spreading on networks.

Overall, Task 2 provides a direct numerical check of the theoretical threshold rule based on $1/\lambda_1$, and the observed behavior above and below the computed thresholds matches the expected outcomes for all the networks considered.

3.3 Conclusion

In this lab, we studied the spreading of a disease using the SIS model on different network topologies. In Task 1, we analyzed the evolution of the proportion of infected nodes over time using a fixed set of model parameters. The results showed that the steady-state infection levels depend strongly on the network structure, with some networks sustaining higher levels of infection than others under the same conditions. The observed differences were consistent with variations in the leading eigenvalue of each network, which captures key structural properties related to connectivity.

In Task 2, we examined how the spreading behavior changes when the model parameters are chosen relative to a network-specific reference value derived from the leading eigenvalue. By selecting infection probabilities slightly above and slightly below this reference value, we were able to compare two distinct regimes of behavior. The results, summarized in the Task 2 table and illustrated in the corresponding figure, show a clear difference between the two cases: infections persist when the infection probability is chosen above the reference value and decrease when it is chosen below it. This behavior was observed consistently across all network types considered.

Overall, the simulations confirm that network structure plays a central role in epidemic spreading. Both tasks highlight the importance of the leading eigenvalue as a useful quantity for characterizing how susceptible a network is to sustained infection in the SIS model. The agreement between simulation results and theoretical expectations supports the relevance of the network-based epidemic framework presented in the course material.

4 Methods

In this study, we implemented a numerical simulation of the Susceptible-Infected-Susceptible (SIS) epidemic model using the Python programming environment. The network generation and manipulation relied on the `networkx` library, while the simulation dynamics and computations were performed using `numpy` and `scipy`.

4.1 SIS Model Implementation

To simulate the spreading dynamics, we developed a function `sis_parallel_step`. Our implementation updates the state of all nodes simultaneously at each time step t based on the system state at $t - 1$. The state transition rules were defined as follows:

1. **Recovery Process:** An infected node i (status 1) recovers to the susceptible state (status 0) with probability γ at each time step.
2. **Infection Process:** A susceptible node i becomes infected if it contracts the disease from any of its infected neighbors. If node i has k_{inf} infected neighbors, the probability of infection $P_{\text{infection}}$ is calculated as:

$$P_{\text{infection}} = 1 - (1 - \beta)^{k_{\text{inf}}} \quad (1)$$

where β is the infection probability per contact. This accounts for independent infection attempts from multiple neighbors.

4.2 Simulation Protocols

The study was conducted in two distinct tasks to analyze both fixed-parameter dynamics and threshold-dependent behaviors. For all simulations, we initialized the networks with a random fraction $p_0 = 0.1$ of infected nodes. To ensure statistical robustness, we performed $N = 100$ independent Monte Carlo iterations for each scenario and computed the mean and standard deviation of the infection prevalence over 50 time steps.

- **Task 1 (Fixed Parameters):** We simulated the epidemic spreading on all network topologies using a fixed infection probability $\beta = 0.15$ and a recovery probability $\gamma = 0.5$. This control experiment allowed for a direct comparison of how topology influences spreading under identical local conditions.
- **Task 2 (Threshold Analysis):** We utilized spectral graph theory to determine network-specific epidemic thresholds. The threshold condition for the SIS model is given by $\frac{\beta}{\gamma} > \frac{1}{\lambda_1}$, where λ_1 is the leading eigenvalue of the adjacency matrix. We computed λ_1 using the `scipy.sparse.linalg.eigsh` function and defined the infection rate as $\beta_c = \gamma/\lambda_1$. We then executed simulations for two regimes:
 - **Super-critical:** $\beta_{\text{high}} = 1.3 \times \beta_c$ (expected survival).
 - **Sub-critical:** $\beta_{\text{low}} = 0.7 \times \beta_c$ (expected extinction).

4.3 Network Topologies

The simulations were performed on five distinct network models, each consisting of $N = 1000$ nodes, to represent various connectivity patterns:

- **Erdős–Rényi (ER):** A random graph generated with edge probability $p = 0.01$, representing a homogeneous random network.
- **Barábasi–Albert (BA):** A scale-free network generated using the preferential attachment mechanism with $m = 5$ edges attached per new node, characterizing heterogeneous degree distributions.
- **Watts–Strogatz (WS):** A small-world network constructed with mean degree $k = 6$ and rewiring probability $p = 0.1$, capturing high clustering coefficients.
- **Complete Graph:** A fully connected network where every node is connected to every other node, approximating the full mixing assumption of classical epidemiology.
- **Star Graph:** A network consisting of one central hub connected to $N - 1$ leaves, representing an extreme case of centralization.